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Code

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Open in... Python 3 (ipykernel)

[38]:
```
pip install pandas scikit-learn matplotlib seaborn
```

```
Requirement already satisfied: pandas in c:\users\user\anaconda3\lib\site-packages (2.1.4)
Requirement already satisfied: scikit-learn in c:\users\user\anaconda3\lib\site-packages (1.2.2)
Requirement already satisfied: matplotlib in c:\users\user\anaconda3\lib\site-packages (3.8.0)
Requirement already satisfied: seaborn in c:\users\user\anaconda3\lib\site-packages (0.12.2)
Requirement already satisfied: numpy<2,>=1.23.2 in c:\users\user\anaconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\user\anaconda3\lib\site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in c:\users\user\anaconda3\lib\site-packages (from pandas) (2023.3.post1)
Requirement already satisfied: tzdata>=2022.1 in c:\users\user\anaconda3\lib\site-packages (from pandas) (2023.3)
Requirement already satisfied: scipy>=1.3.2 in c:\users\user\anaconda3\lib\site-packages (from scikit-learn) (1.11.4)
Requirement already satisfied: joblib>=1.1.1 in c:\users\user\anaconda3\lib\site-packages (from scikit-learn) (1.2.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\user\anaconda3\lib\site-packages (from scikit-learn) (2.2.0)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (23.1)
Requirement already satisfied: pillow>=6.2.0 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (10.2.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\user\anaconda3\lib\site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: six>=1.5 in c:\users\user\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

[39]:
```
import pandas as pd

# Load the dataset
df = pd.read_csv('tested.csv')
```

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[39]:
```python
import pandas as pd

# Load the dataset
df = pd.read_csv('tested.csv')

# Display the first few rows
print(df.head())
```

```
   PassengerId  Survived  Pclass  \
0          892         0       3
1          893         1       3
2          894         0       2
3          895         0       3
4          896         1       3

                                           Name     Sex   Age  SibSp  Parch  \
0                              Kelly, Mr. James    male  34.5      0      0
1              Wilkes, Mrs. James (Ellen Needs)  female  47.0      1      0
2                     Myles, Mr. Thomas Francis    male  62.0      0      0
3                              Wirz, Mr. Albert    male  27.0      0      0
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1      1

    Ticket     Fare Cabin Embarked
0   330911   7.8292   NaN        Q
1   363272   7.0000   NaN        S
2   240276   9.6875   NaN        Q
3   315154   8.6625   NaN        S
4  3101298  12.2875   NaN        S
```

[40]:
```python
df.tail(5)
```

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Code

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[40]:
```
df.tail(5)
```

[40]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
413	1305	0	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500	NaN	S
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000	C105	C
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	1308	0	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500	NaN	S
417	1309	0	3	Peter, Master. Michael J	male	NaN	1	1	2668	22.3583	NaN	C

[41]:
```
df.dtypes
```

[41]:
```
PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
dtype: object
```

[42]:
```
"""age, gender, ticket class, fare, cabin, and whether or not they survived """
```

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Code · Open in... Python 3 (ipykernel)

[42]:
```
"""age, gender, ticket class, fare, cabin, and whether or not they survived."""
df=df.drop(['Name', 'SibSp', 'Parch', 'Embarked'], axis=1)
df.head(5)
```

[42]:

	PassengerId	Survived	Pclass	Sex	Age	Ticket	Fare	Cabin
0	892	0	3	male	34.5	330911	7.8292	NaN
1	893	1	3	female	47.0	363272	7.0000	NaN
2	894	0	2	male	62.0	240276	9.6875	NaN
3	895	0	3	male	27.0	315154	8.6625	NaN
4	896	1	3	female	22.0	3101298	12.2875	NaN

[43]:
```
df.shape
```

[43]:
```
(418, 8)
```

[44]:
```
duplicate_rows_df = df[df.duplicated()]
print("number of duplicate rows: ", duplicate_rows_df.shape)
```

```
number of duplicate rows:  (0, 8)
```

[45]:
```
df.count()
```

[45]:
```
PassengerId    418
Survived       418
Pclass         418
Sex            418
```

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```
dtype: int64
```

[46]:
```
df = df.drop_duplicates()
df.head(5)
```

[46]:

	PassengerId	Survived	Pclass	Sex	Age	Ticket	Fare	Cabin
0	892	0	3	male	34.5	330911	7.8292	NaN
1	893	1	3	female	47.0	363272	7.0000	NaN
2	894	0	2	male	62.0	240276	9.6875	NaN
3	895	0	3	male	27.0	315154	8.6625	NaN
4	896	1	3	female	22.0	3101298	12.2875	NaN

[47]:
```
df.count()
```

[47]:
```
PassengerId    418
Survived       418
Pclass         418
Sex            418
Age            332
Ticket         418
Fare           417
Cabin           91
dtype: int64
```

[48]:
```
print(df.isnull().sum())
```

```
PassengerId     0
Survived        0
```

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[49]:
```
df = df.dropna()    # Dropping the missing values.
df.count()
```

[49]:
```
PassengerId    87
Survived       87
Pclass         87
Sex            87
Age            87
Ticket         87
Fare           87
Cabin          87
dtype: int64
```

[50]:
```
print(df.isnull().sum())    # After dropping the values
```

```
PassengerId    0
Survived       0
Pclass         0
Sex            0
Age            0
Ticket         0
Fare           0
Cabin          0
dtype: int64
```

[51]:
```
df.head(5)
```

[51]:

	PassengerId	Survived	Pclass	Sex	Age	Ticket	Fare	Cabin
12	904	1	1	female	23.0	21228	82.2667	B45
14	906	1	1	female	47.0	W.E.P. 5734	61.1750	E31

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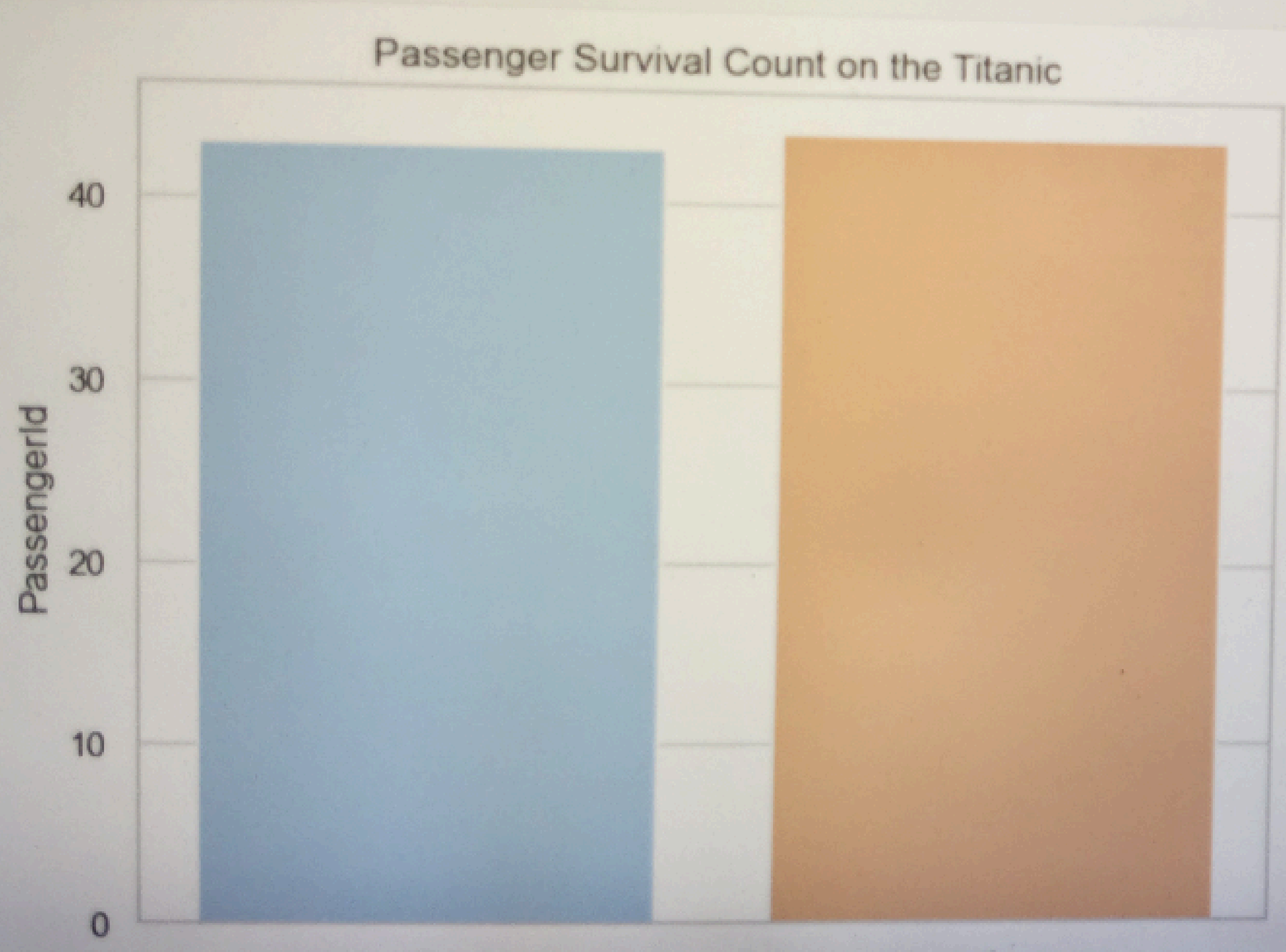
[55]:
```
sns.set(style="whitegrid")
sns.countplot(x='Survived', data=df, palette='pastel')
plt.title('Passenger Survival Count on the Titanic')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('PassengerId')
plt.show()
```

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Python 3 (ipykernel)
Passenger Survival Count on the Titanic
PassengerId
0
10
20
30
40
0
1
Survived (0 = No, 1 = Yes)
[ ]: